



SETTING UP A CLOUD

IN AN ENGINEERING COLLEGE

Cloud computing, the delivery of computing as a service rather than as a product, offers shared resources, software, and information to computers and other devices as a service over a network. Amazon's EC2, Microsoft's Azure, Google's AppEngine and IBM's Blue Cloud are examples. There are three kinds of service modes—Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). From the deployer's perspective, cloud computing platforms may be private (within an organisation), public, or hybrid (a combination of private and public). Even governments of countries like the US, UK and Singapore have started investing in cloud computing. This article describes how the Thiagarajar College of Engineering in Tamil Nadu went about setting up a cloud environment in its premises.

Computing power is needed in almost all fields today, from business to medicine. And in the academic community, computers are used by members of almost all disciplines. Certain fields of research require multiple cores, gigabytes of RAM, and possibly, terabytes of storage. It would be expensive or infeasible to provide such physical resources to each user; this issue is addressed by IaaS cloud computing platforms that provide on-demand virtual machines to users. After users terminate these virtual machines, resources are freed up for the next user. On a LAN, this service will usually be seamless, due to the low latency.

Building a cloud

In July 2010, Rackspace and NASA jointly launched a new open source cloud initiative known as OpenStack, which aimed at allowing any organisation to create and offer cloud-based services. It primarily provided IaaS. More than 120 prominent companies such as Dell, AMD, Intel, HP and Cisco have joined this project. OpenStack integrates code from NASA's Nebula and Rackspace's Cloud Files platforms. It has a modular architecture comprising three components: Compute (Nova), Object Storage (Swift) and Image Service (Glance). These include services such as Nova Scheduler, Nova API (provides